

What effects our happiness?

Final Project Report

Group 40

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1 Introduction and problem description

1.1 Background and Context

In an increasingly interconnected world, understanding the well-being of nations has become an important factor in shaping global economic, social, and policy decisions. Indicators such as happiness, employment, and economic performance have become vital metrics for assessing the quality of life in countries across the globe. This report aims to investigate the interplay between these variables to provide a comprehensive view of their collective influence on societal progress.

Happiness is a multidimensional concept encompassing emotional well-being, societal support, and life satisfaction. The World Happiness Report is published annually, and provides valuable insights into how countries rank in terms of happiness based on economic, social, and cultural factors¹. Similarly, employment levels which reflect economic stability and opportunities, serve as critical indicators of societal welfare. Unemployment rates often highlight economic disparities and their implications on mental health, productivity, and societal cohesion. Lastly, economic data, such as GDP per capita, offers insight into the material well-being of populations, further contextualizing trends in happiness and employment.

1.2 Objective of the Study

The objective of this report is to synthesize data from various global datasets to uncover patterns and relationships among happiness, employment, and economic metrics. By leveraging datasets such as the World Happiness Report, global unemployment statistics, and GDP-related indicators, this study aims to provide an integrated analysis that highlights the underlying factors driving societal well-being.

Specifically, the report seeks to answer the following key questions:

1. How does happiness correlate with economic indicators like GDP per capita?
2. To what extent does employment impact happiness levels?
3. Are there significant regional or cultural differences in the relationships between these factors?

By addressing these questions, the report hopes to provide useful observations that contribute to a better understanding of the factors influencing happiness across different contexts.

1.3 Scope and Methodology

The analysis in this report is built on datasets curated from reliable sources available on Kaggle, including:

- **World Happiness Report (2015–2019):** Measures happiness scores across countries based on variables like social support, income, and freedom.

¹ <https://worldhappiness.report/about/>

- **Global Employment Data:** Provides insights into unemployment trends by region and demographic.
- **Economic Performance Data:** Covers metrics such as GDP growth and GDP per capita to offer a robust picture of economic conditions.

The report employs data-cleaning techniques to ensure consistency and reliability, followed by merging and analyzing these datasets to identify trends and correlations. Advanced statistical methods and visualizations are used to distill complex relationships into actionable insights.

1.4 Importance of the Study

This study provides valuable insights into the relationship between happiness, employment, and economic metrics by analyzing publicly available data. The findings aim to uncover patterns and correlations within the data that contribute to a deeper understanding of these complex socioeconomic factors. By utilizing statistical methods, this study not only enhances our knowledge of how happiness and employment are influenced by economic conditions but also highlights the importance of grounding quantitative analysis within broader societal contexts. Through this process, the study encourages critical thinking about data-driven decision-making and the potential for such analyses to inform policy, business, and research in the future.

1.5 Structure of the Report

This report is organized into the following sections:

- **Description of the data**
- **Methodology**
- **Results and analysis.**

By integrating these components, the report aspires to deliver a nuanced understanding of the factors that define and influence global well-being.

2 Description of the data

2.1 Data sources and Selection criteria

As previously mentioned, this project utilizes three primary datasets: the World Happiness Score, global economic data, and global employment data. These datasets were not chosen at random, but rather selected due to an interest in examining correlations between economic and employment factors and their effects on the World Happiness Index.

2.1.1 World Happiness Index

First, we describe the factors that contribute to the values presented in the World Happiness Report. The World Happiness Report collaborates with Gallup² to collect data used in calculating the indices. Gallup conducts World Poll Surveys, and the World Happiness Report uses life evaluation data from these surveys to determine the global happiness index. While the Gallup Poll varies from year to year, in 2024 respondents were asked to rate their lives on a scale from 0 to 10.

According to the report, six key factors contribute most significantly to the happiness rating:

- GDP
- Life expectancy
- Generosity
- Social support
- Freedom
- Corruption

Upon reviewing the 2024 report, several intriguing patterns emerged. Countries from the same region sometimes clustered together in the rankings, while at other times they were more dispersed. We aimed to determine whether there is a correlation between these contributing factors and the region, or if the factors influence happiness scores similarly regardless of regional context.

Another notable finding, which we did not prioritize in this project, relates to countries whose high happiness rankings were unexpected. For instance, Israel ranked as the 5th happiest country globally despite having been involved in the Israel-Hamas conflict for more than a year³. Intuitively, one might expect war to negatively impact a country's happiness. In this case, however, the data defied that expectation. While it would be interesting to investigate this phenomenon further, it fell outside the scope of our current analysis.

2.1.2 World Economic Data

Our primary goal was to understand how specific features influence the happiness index. We chose features that are somewhat interrelated yet distinct enough to provide meaningful comparisons.

We selected two economic datasets: GDP and GDP growth. The World Happiness Report suggests that GDP has a strong correlation with a country's happiness score. We included this feature to verify whether the relationships proposed in the World Happiness Report hold true, and to see if our machine learning models would also identify these correlations.

Of particular interest was GDP growth. We wanted to explore whether changes in GDP over time would correlate strongly with happiness. This question will be examined further in the [hypothesis section](#).

² <https://worldhappiness.report/faq/>

³ <https://www.nytimes.com/news-event/israel-hamas-gaza>

2.1.3 World Employment Data

We also wanted to consider a variable that, while related, might still provide different insights. Unemployment rates can correlate with GDP, but they also possess unique characteristics. Our intent was to investigate whether unemployment rates, despite their relationship to GDP, would independently show a significant correlation with the happiness index.

2.1.4 World Region Data

We additionally incorporated a dataset that maps countries to their respective regions. Although we do not conduct any direct analysis on this dataset, it is used to facilitate regional comparisons and mappings.

2.1.5 Other considerations

We considered integrating other datasets that could also be informative. However, we ultimately chose to focus on the selected data that we found most compelling. Adding more features could indeed offer a broader understanding of the factors contributing to the World Happiness Index. Some noteworthy datasets we considered but did not include are:

- Carbon footprint
- Crime index
- Cost of living
- Alcohol consumption
- Pizza consumption

While these additional factors might be intriguing to analyze, we preferred to concentrate our efforts on the chosen datasets to gain a deeper understanding of the key variables.

2.2 Data Acquisition, Format and Features

2.2.1 Acquisition

All of our data was sourced from Kaggle. The data was downloaded using Kaggle's API. The code for downloading and organizing the datasets can be found in the accompanying Jupyter notebook titled `data_preprocessing`. Section 1 of that notebook details the steps taken to download the data from Kaggle.

2.2.2 Format and Features

The datasets were originally provided as separate annual files and were not combined. The process of integrating and preparing the relevant features is demonstrated in Section 2 of the `data_preprocessing` notebook. The final integrated dataset includes annual data from 2015 to 2019 for the following variables: Happiness Score, Unemployment Rate, GDP per capita, and GDP Growth per capita. We also calculated the mean values for each of these indicators across all years.

Country	Region	HS	HS_Mean	UR	UR_Mean	G	G_Mean	GG	GG_Mean
Norway	...								
...									

HS = Happiness score, UR = Unemployment rate, G = GDP per capita, GG = GDP growth per capita

2.3 Data quality and cleaning

The datasets had no missing values (see data_analysis, Section 0), so no additional cleaning was required. However, some challenges arose when merging datasets by country (see data_preprocessing, Section 3), causing us to lose some data points. This limitation is worth noting, as the global sample size is inherently small. Losing even a few data points can be meaningful when building and training models with a limited number of countries.

3 Methodology

3.1 Hypothesis

Before conducting the analysis, we wish to establish clear hypotheses based on the observed patterns discussed in the previous sections. Drawing on the background provided by the World Happiness Report, as well as the data characteristics highlighted in the Data Description section, we propose the following hypotheses:

3.1.1 Happiness and GDP & GDP Growth

H1: Countries with higher GDP & GDP Growth per capita will exhibit higher happiness.

The World Happiness Report indicates GDP as one of the most influential factors in determining happiness. We therefore expect a positive correlation between GDP per capita and the overall happiness index.

3.1.2 Happiness and Unemployment:

H2: Higher unemployment rates will correlate negatively with happiness scores.

Although unemployment and GDP are related, unemployment adds another dimension to economic well-being—job security and the ability to earn a stable income. Reduced employment opportunities often lead to financial strain, social stress, and lower life satisfaction, all of which are likely to decrease happiness.

3.1.3 Regional Variations:

H3: The impact of factors on happiness will vary across regions.

Cultural, social, and political factors may shape how economic conditions influence perceptions of happiness. We anticipate that certain regions may demonstrate stronger or weaker correlations among these variables due to differing cultures.

3.2 Statistical and Analytical Techniques

To investigate the relationships between happiness, GDP, GDP growth, and unemployment across various regions, we employ a combination of statistical and machine learning techniques. Each technique brings unique insights and strengths to the analysis, allowing us to both understand and model the underlying patterns in our data.

3.2.1 Descriptive Statistics (data_analysis, section 1)

Before delving into complex models, it is essential to gain a foundational understanding of our data. By examining the mean, median, and standard deviation of our variables, we obtain a clear overview of the data. This initial step helps identify outliers, potential data quality issues. After examining the descriptive statistics we decided that our data quality was up to standard.

3.2.2 Correlation (data_analysis, section 2 and 7)

By examining correlation coefficients between happiness and each economic and employment indicator, we can determine whether these factors move together or not, and more importantly how significant each of the factors correlate with the happiness score. This technique is also used to analyse the importance of factors in different regions.

3.2.3 Linear Regression (data_analysis, section 3)

Both simple linear regression and multiple linear regression is used. Linear regression models trained on the data were used to visualize how accurately the happiness score could be predicted and whether it was easier to predict lower happiness scores compared to higher scores.

3.2.4 Neural Network (data_analysis, section 4)

Neural networks are powerful, flexible models capable of capturing complex, nonlinear relationships in the data. While linear regression assumes a straight-line relationship, neural networks can adapt to a wide range of patterns. Our main goal behind using this model was most for our own interest, we wanted to find out if it could perform on such a small dataset.

3.2.5 Decision Tree (data_analysis, section 5)

Decision trees offer a more interpretable approach to modeling complex relationships. By recursively splitting the data based on feature values, decision trees highlight which variables have the most significant impact on happiness, which is exactly what we wanted to analyse.

3.2.6 Random Forest (data_analysis, section 6)

Random forests, an ensemble method that aggregates multiple decision trees, enhance model robustness and generalization. By averaging predictions across a diverse set of trees, random forests reduce the risk of overfitting and often improve predictive accuracy. And we wanted to see if the added investment of generating random forest compared to a single decision tree would provide significant difference.

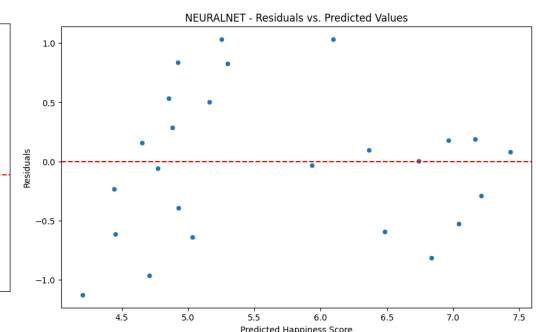
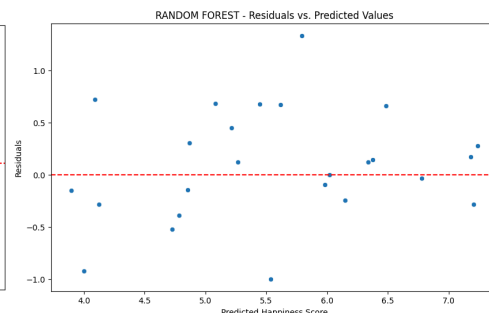
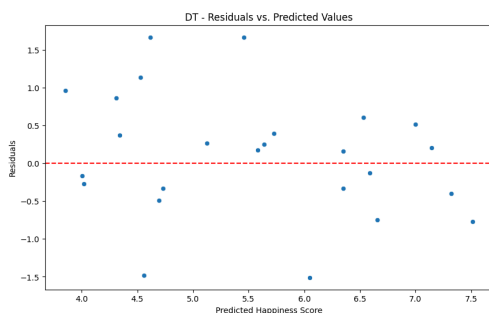
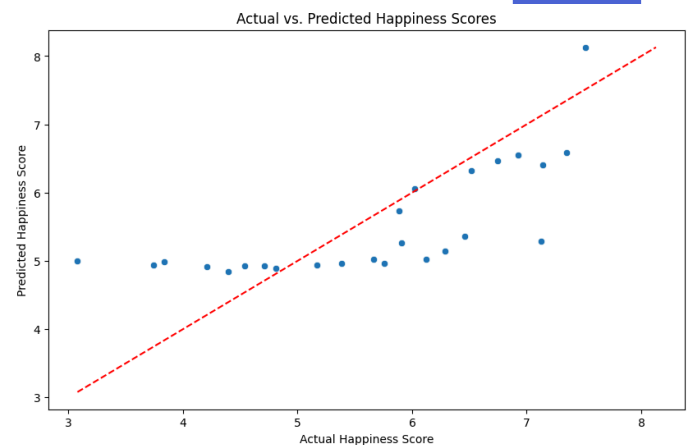
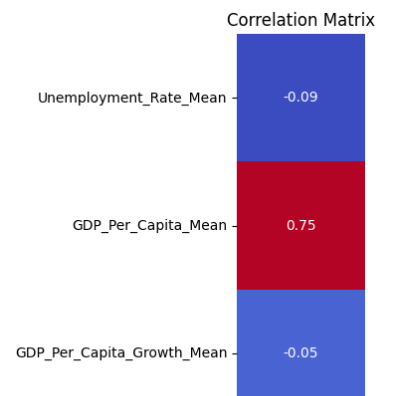
4. Results

4.1 Findings: Features and happiness (data_analysis, section 1-6)

The correlation matrix revealed that GDP per capita had a notable positive correlation with the happiness scores of countries. However, unemployment rate and GDP growth showed no significant correlation, contradicting our initial hypothesis.

We then trained a linear regression model to predict happiness scores based on these features. The model performed poorly at predicting lower happiness scores but showed better accuracy for higher scores. This suggests that the selected features (GDP per capita, GDP growth, and unemployment rate) may have a stronger influence on happiness in countries that already rank highly on the happiness index.

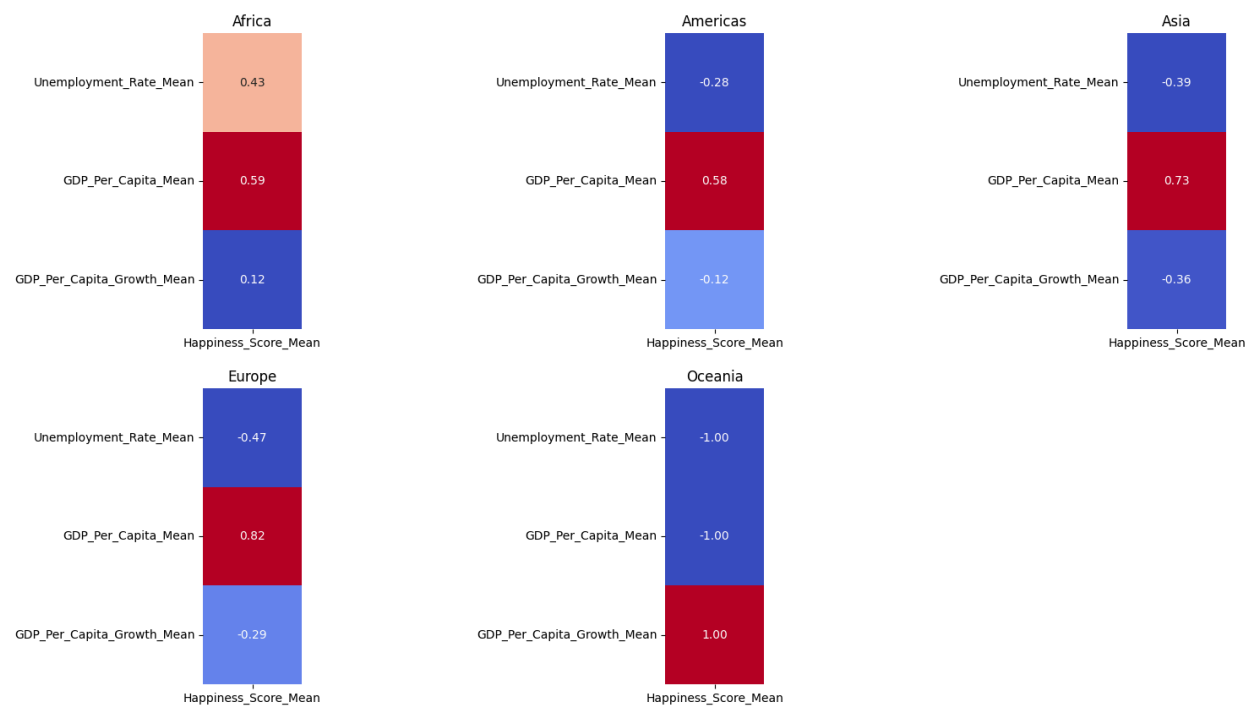
To explore whether alternative models might yield better results, we tested a neural network, a decision tree, and a random forest. Residual analysis of these models indicated that none performed significantly better than linear regression. The lack of predictive power suggests that our dataset and feature selection were insufficient for reliably modeling happiness scores. With a larger dataset or additional relevant features, it might be possible to achieve better predictive accuracy. Consequently, we were unable to confirm hypotheses [H1](#) or [H2](#).



4.2 Findings: Regional differences (data_analysis, section 7)

Our regional analysis revealed significant variations in the factors influencing happiness. It is important to note that dividing the dataset into regions resulted in smaller sample sizes, which could affect the robustness of these findings.

In Europe, GDP per capita showed a much stronger correlation with happiness compared to other regions. Interestingly, unemployment rate exhibited a positive correlation with happiness in Africa, an unexpected result. These findings align with our hypothesis [H3](#), highlighting how cultural, economic, and social contexts shape the relationship between happiness and its influencing factors.



5. Reflection

This project provided valuable insights into the challenges of applying machine learning to real-world data, particularly the importance of having sufficient and high-quality datasets. The limited scope of our data, constrained by the fact that there is a finite number of countries, impacted the reliability and performance of our models.

However, we still uncovered interesting findings, such as regional variations in the factors influencing happiness, which highlighted the importance of context in data analysis. This experience emphasized the need for careful feature selection and demonstrated how larger or more comprehensive datasets could lead to better results.